

Length contraction:

$$L = \gamma^{-1} L_0 < L_0$$

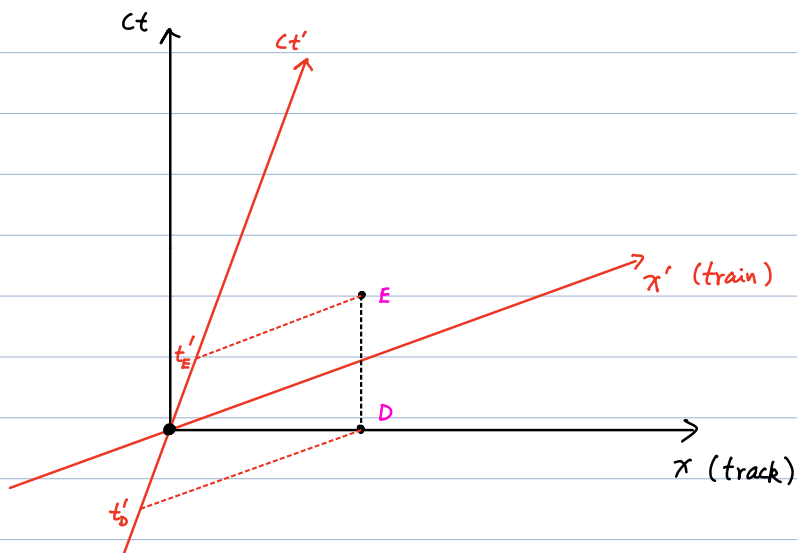
Time dilation:

$$\Delta t = \gamma \Delta t_0 > \Delta t_0$$

Puzzle:

| Observer |                                   |
|----------|-----------------------------------|
| track    | length of <i>train</i> contracted |
|          | clock in <i>train</i> slowed down |
| train    | length of <i>track</i> contracted |
|          | clock on <i>track</i> slowed down |

Contradiction? No!



clock stationary in K, expect:

$$\Delta t'_{DE} = \gamma \Delta t_{DE}$$

$\downarrow$   
 $\Delta t_0$

Proof:

$$\begin{pmatrix} ct'_E \\ x'_E \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} ct_E \\ x_E \end{pmatrix} = \begin{pmatrix} \gamma ct_E - \beta\gamma x_E \\ -\beta\gamma ct_E + \gamma x_E \end{pmatrix}$$

①

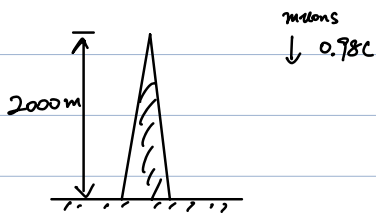
$$\begin{pmatrix} ct'_b \\ x'_b \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} 0 \\ x_E \end{pmatrix} = \begin{pmatrix} -\beta\gamma x_E \\ \gamma x_E \end{pmatrix}$$

$$\Rightarrow \Delta t'_{DE} = \gamma t_E = \gamma \Delta t_{DE} \quad \text{proved.}$$

• Example. Muon decay

Half-life at rest:  $t_{1/2} \approx 1.52 \mu\text{s}$

$$N = N_0 2^{-t/t_{1/2}} \quad (t = t_{1/2} \Rightarrow N = \frac{1}{2} N_0)$$



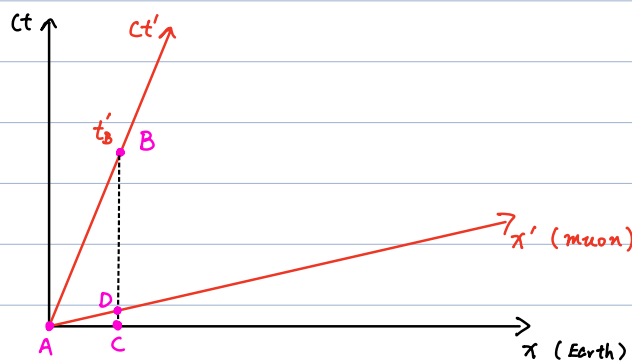
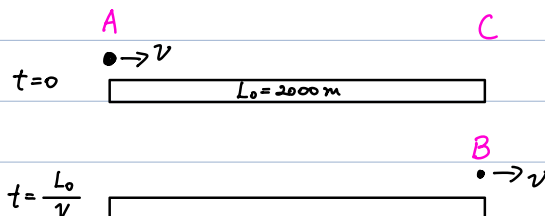
Q: how many reach surface?

Naively.

$$t = \frac{2000 \text{ m}}{0.98c} \approx 6.8 \mu\text{s}$$

$$\Rightarrow \frac{N}{N_0} \approx 2^{-6.8/1.52} \approx 4.5\% \quad \text{reach surface}$$

Relativity:



A: start

B: reach surface

① Use earth frame:

A → B (surface stationary)

$$\begin{cases} x_B = L_0 \\ t_B = 6.8 \mu s = \frac{L_0}{v} \end{cases} \quad v = 0.98c$$

$$\begin{pmatrix} ct'_B \\ x'_B \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} ct_B \\ L_0 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\gamma}{\beta} L_0 - \beta\gamma L_0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{L_0}{\beta\gamma} \\ 0 \end{pmatrix}$$

time passed in the muon frame:

$$\Delta t' = t'_B = \frac{L_0}{v\gamma} = \gamma^{-1} t_B \quad \text{time dilation!}$$

$$\approx 1.35 \mu s$$

$$\Rightarrow \frac{N}{N_0} \approx 2^{-1.35/1.52} \approx 54\%$$

In other words, earth observer finds

$$t_{1/2}^{(\text{earth})} = \gamma t_{1/2} \approx 7.6 \mu s \quad (\text{time dilation})$$

② Use muon frame:

$$\begin{array}{ccc} C & \longrightarrow & D \longrightarrow B \\ \downarrow & & \downarrow \\ t' < 0 & & t' = 0 \end{array} \quad (\text{muon stationary})$$

$$x'_D - x'_B = \gamma^{-1} L_0 \approx 398 \text{ m} \quad (\text{length contraction})$$

proof:  $x'_B = 0$  (see above)

$$\begin{pmatrix} ct'_D \\ x'_D \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} ct_D \\ x_D \end{pmatrix} = \begin{pmatrix} \gamma ct_D - \beta\gamma L_0 \\ -\beta\gamma ct_D + \gamma L_0 \end{pmatrix}$$

$$t'_D = 0 \Rightarrow ct_D = \beta L_0$$

$$\Rightarrow x'_D = -\beta\gamma \cdot \beta L_0 + \gamma L_0$$

$$= \gamma L_0 (1 - \beta^2)$$

$$= \gamma^{-1} L_0 \quad \text{proved.}$$

⇒ time need to go from D → B:

③

$$t'_{DB} = \frac{\gamma^{-1} L_0}{v} \approx 1.35 \text{ } \mu\text{s}$$

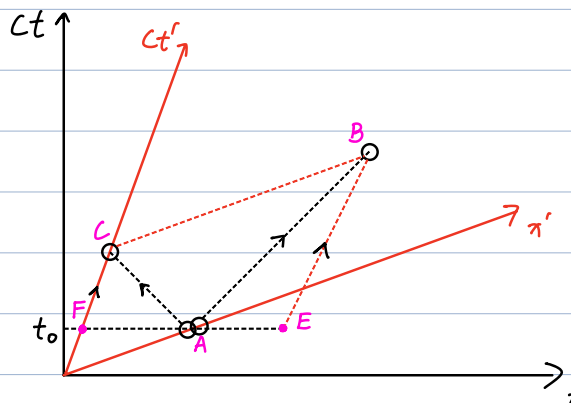
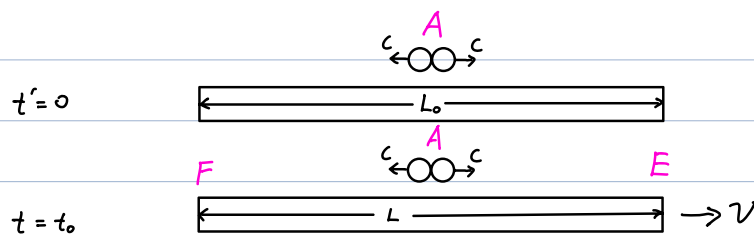
$$\Rightarrow \frac{N}{N_0} \approx 2^{-1.35/1.52} \approx 54\%$$

Summary.

|           | Earth frame                                       | Muon frame                                                    |
|-----------|---------------------------------------------------|---------------------------------------------------------------|
| distance  | $L_0 = 2000 \text{ m}$                            | $\gamma^{-1} L_0 \approx 398 \text{ m}$                       |
| half-life | $\gamma t_{1/2} \approx 7.6 \text{ } \mu\text{s}$ | $t_{1/2} \approx 1.52 \text{ } \mu\text{s}$                   |
| time      | $\frac{L_0}{v} \approx 6.8 \text{ } \mu\text{s}$  | $\frac{\gamma^{-1} L_0}{v} \approx 1.35 \text{ } \mu\text{s}$ |

• Example

Q: train front & rear locations in K frame at  $t = t_0$ ?



$$A: t'_A = 0, x'_A = \frac{L_0}{2}$$

$$\Rightarrow \begin{pmatrix} ct_0 \\ x_A \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} 0 \\ \frac{L_0}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2}\beta\gamma L_0 \\ \frac{1}{2}\gamma L_0 \end{pmatrix}$$

$$\Rightarrow x_A = \frac{\gamma}{2} L_0 \quad (\text{NOT } \frac{\gamma^{-1}}{2} L_0 \text{ !})$$

Method-1. Lorentz TR :

$$F: ct_0 = \frac{1}{2} \beta \gamma L_0$$

$$\Rightarrow x_F = v t_0 = \frac{1}{2} \beta^2 \gamma L_0$$

$$x_A - x_F = (1 - \beta^2) \gamma \frac{L_0}{2} = \gamma^{-1} \frac{L_0}{2}$$

$\Rightarrow$  consistent with length contraction !

Method-2. length contraction

$$x_F = x_A - \gamma^{-1} \frac{L_0}{2}$$

$$= \frac{1}{2} \beta^2 \gamma L_0$$

### Invariant interval

• Galilean transformation :

$$(\Delta s)^2 \equiv (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

invariant !

• Lorentz transformation

$$(\Delta s)^2 \equiv c^2 (\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2$$

invariant !

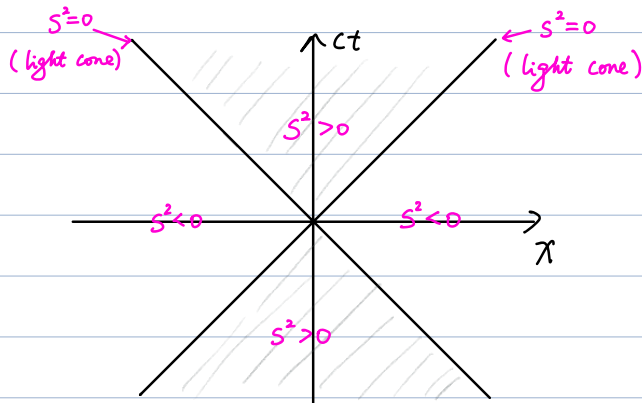
$$\begin{pmatrix} ct \\ x \end{pmatrix} = \begin{pmatrix} \gamma & \beta \gamma \\ \beta \gamma & \gamma \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix}$$

$$\Rightarrow c^2 t^2 - x^2 = (\gamma c t' + \beta \gamma x')^2 - (\beta \gamma c t' + \gamma x')^2$$

$$= (1 - \beta^2) \gamma^2 c^2 t'^2 + (\beta^2 - 1) \gamma^2 x'^2$$

$$= c^2 t'^2 - x'^2$$

proved.



- Time like separation (类时间隔):  $s^2 > 0$

- possible to find a frame where  $x=0$  (2 events at same place)

- proper time:  $s/c$  (固有时间)

- Space like separation (类空间隔):  $s^2 < 0$

- possible to find a frame where  $t=0$  (2 events at same time)

- proper length (distance):  $s$  (固有长度)

- Light like separation (类光间隔):  $s^2 = 0$

- causality (因果律)